



Introduction to Cloud Infrastructure [AZ-900]

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Date



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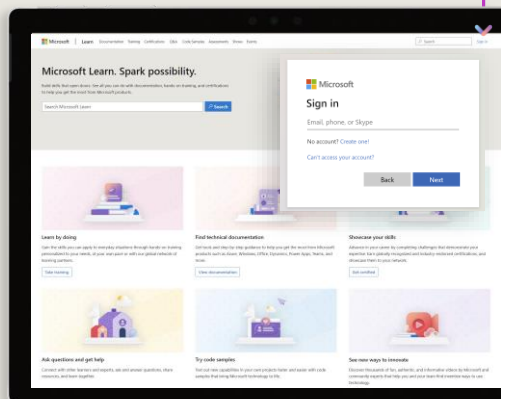
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This course includes labs:

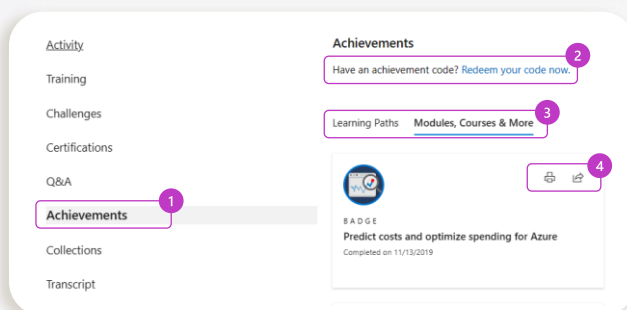
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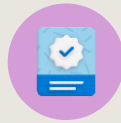


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¹ "2023 Value of IT Certification | Candidate Report," Pearson VUE, 2023 ² Microsoft fundamentals certifications don't expire

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Get ready for your Microsoft Certification exam

Microsoft Certified: Azure Fundamentals

Understand the skills measured by the exam

Study area	Percentage
Describe cloud concepts	25-30%
Describe Azure architecture and services	35-40%
Describe Azure management and governance	30-35%

Percentages indicate the relative weight of each area on the exam.
The higher the percentage, the more questions you are likely to see in that area.

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Build confidence in your skills



In the exam page, find resources to help prepare

- Watch exam prep videos
- Review the exam study guide
- Demo the exam experience with the exam sandbox
- Take a practice assessment

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Cloud Concepts

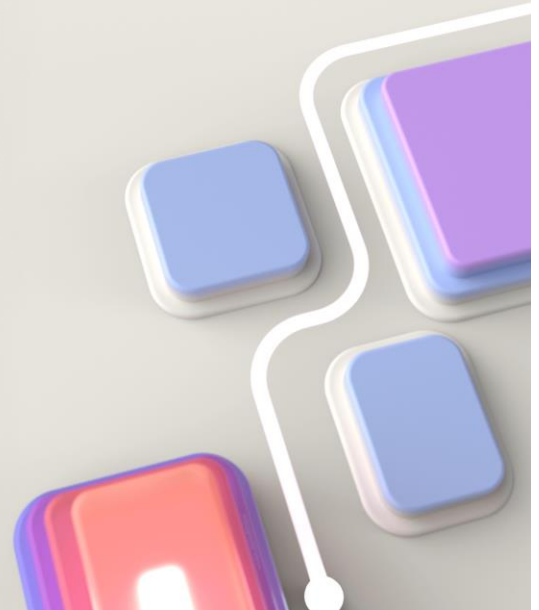
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Describe cloud computing

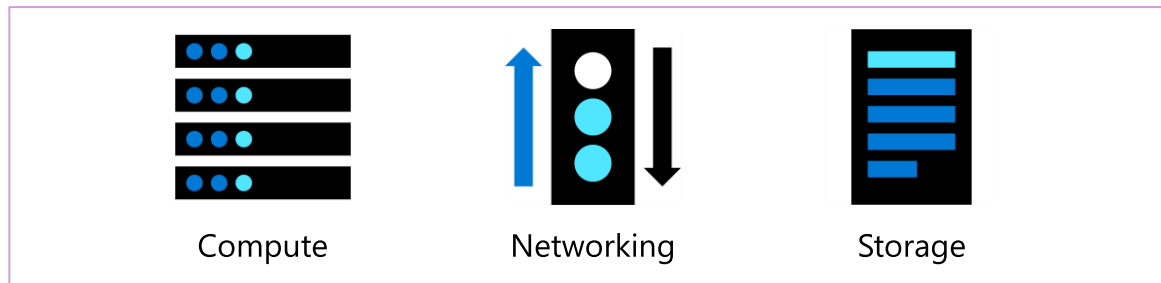
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What is cloud computing?

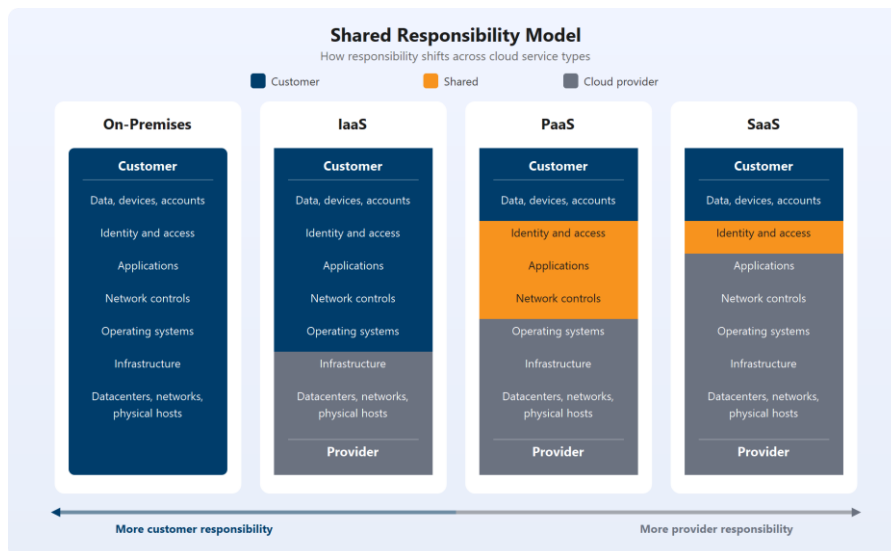
Cloud computing is the delivery of computing services over the internet, enabling faster innovation, flexible resources, and economies of scale.



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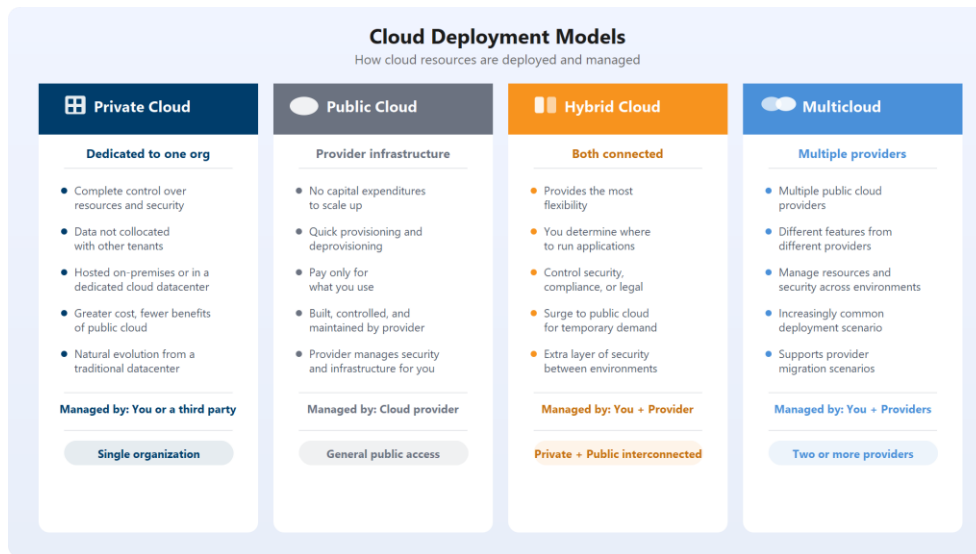
The shared responsibility model



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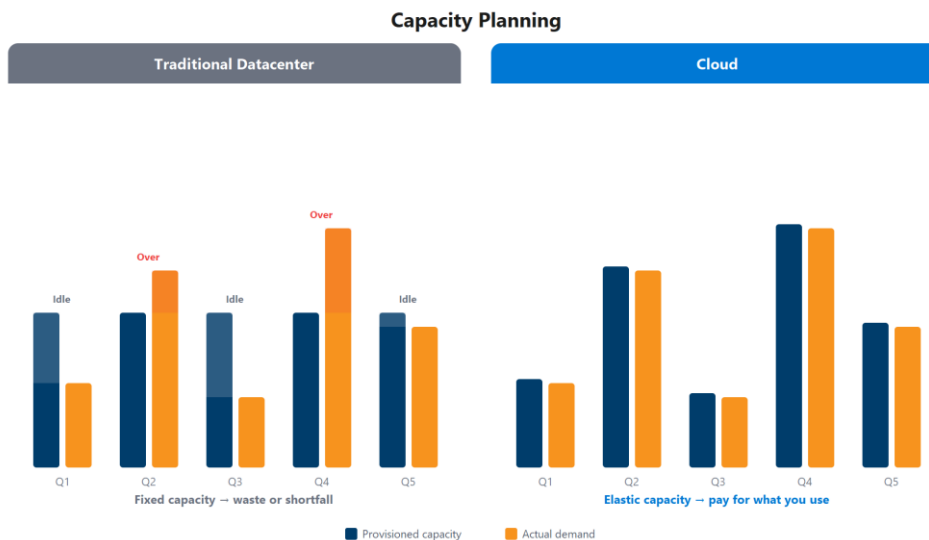
Cloud deployment models



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The consumption-based model

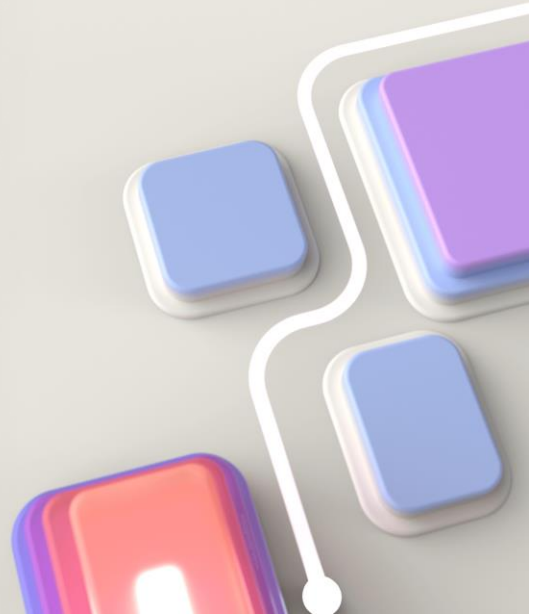


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Describe the benefits of using cloud services

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High availability and scalability

High availability

- Focus on maximum availability regardless of disruptions or events
- Azure SLAs define uptime guarantees for each service
- Architect solutions with high availability in mind

Scalability

- Adjust resources to match changing demand
- Vertical scaling: increase or decrease CPU/RAM of existing resources
- Horizontal scaling: add or subtract the number of resources

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Reliability and predictability

Reliability

- Ability of a system to recover from failures and continue to function
- Part of the Microsoft Azure Well-Architected Framework
- Decentralized design enables deployment in regions around the world
- Automatic failover to a different region in the event of catastrophe

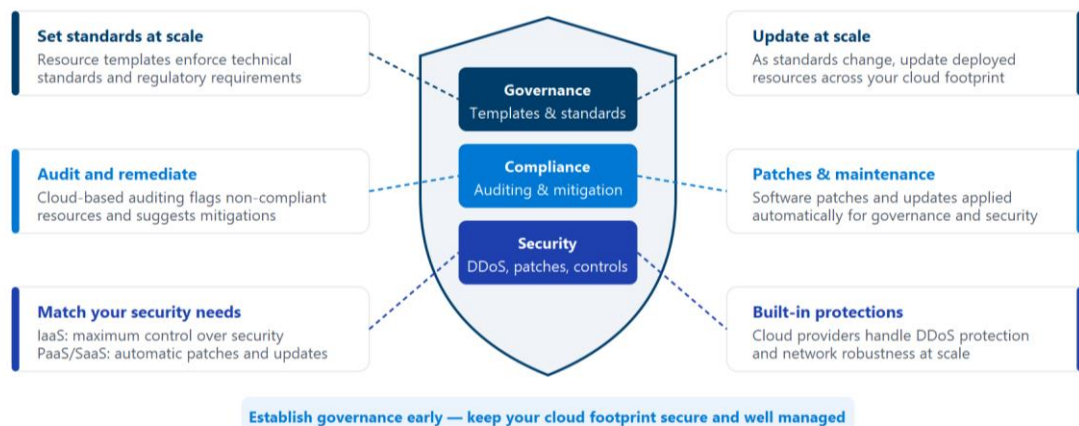
Predictability

- Performance predictability: predict resources needed for a good experience
- Cost predictability: predict or forecast the cost of cloud spend
- Azure Pricing Calculator helps estimate cloud cost

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Security and governance in the cloud



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Manageability in the cloud

Management of the cloud

- Automatically scale resource deployment based on need
- Deploy resources based on preconfigured templates
- Monitor the health of resources and automatically replace failures
- Receive automatic alerts based on configured metrics

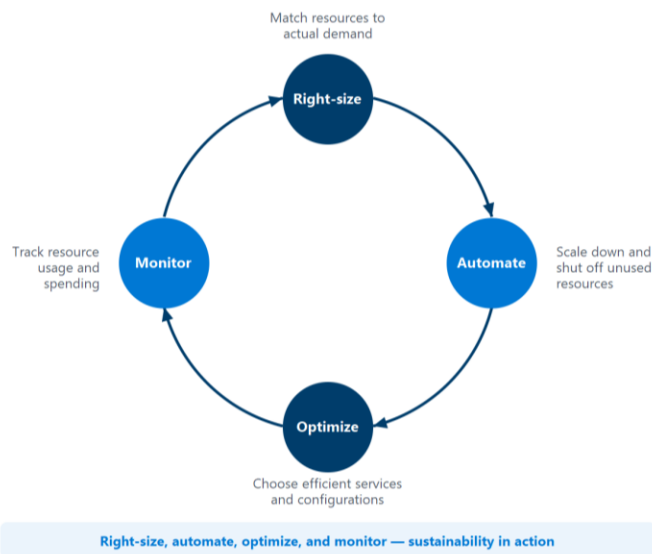
Management in the cloud

- Through a web portal
- Using a command-line interface (CLI)
- Using APIs
- Using PowerShell

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Sustainability in the cloud



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Describe cloud service types

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Cloud service types comparison

IaaS Most control	PaaS Balanced	SaaS Least control
Applications	Applications	Applications
Data	Data	Data
Runtime	Runtime	Runtime
Middleware	Middleware	Middleware
Operating System	Operating System	Operating System
Servers	Servers	Servers
Storage	Storage	Storage
Networking	Networking	Networking

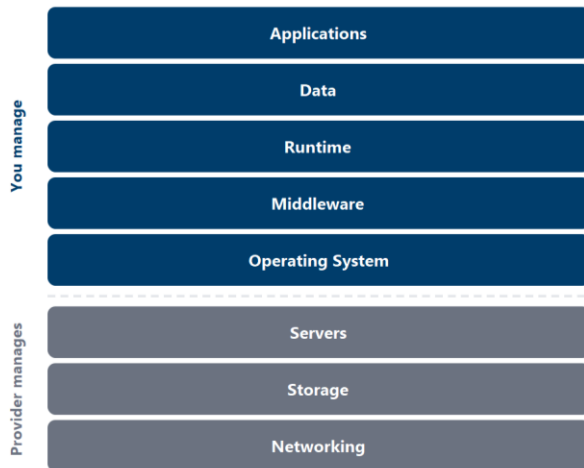
■ You manage
 ■ Cloud provider manages

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Infrastructure as a service (IaaS)

Maximum control — you manage the workload stack



Common Scenarios

Lift-and-shift migration

Set up cloud resources similar to on-premises, then move workloads to IaaS infrastructure.

Testing and development

Rapidly replicate dev/test environments. Start up or shut down with complete control.

Key takeaway

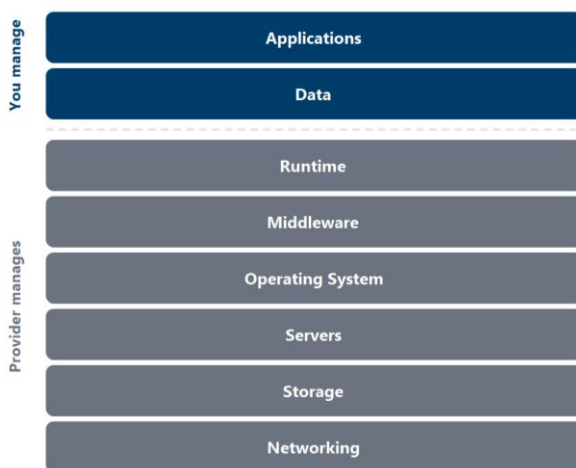
You rent hardware — what you do with it is up to you.

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Platform as a service (PaaS)

Balanced — you focus on your application and data



Common Scenarios

Development framework

Build cloud-based applications using built-in components with scalability and high availability.

Analytics and business intelligence

Analyze and mine data, find insights, and predict outcomes using platform tools.

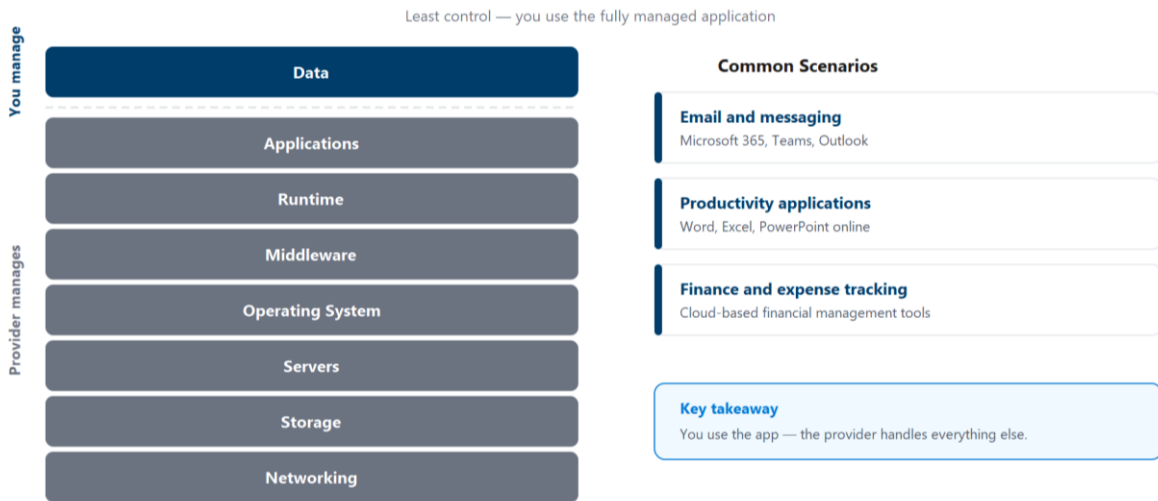
Key takeaway

You build on the platform — no infrastructure headaches.

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Software as a service (SaaS)



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Azure architecture and services



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Describe the core architectural components of Azure

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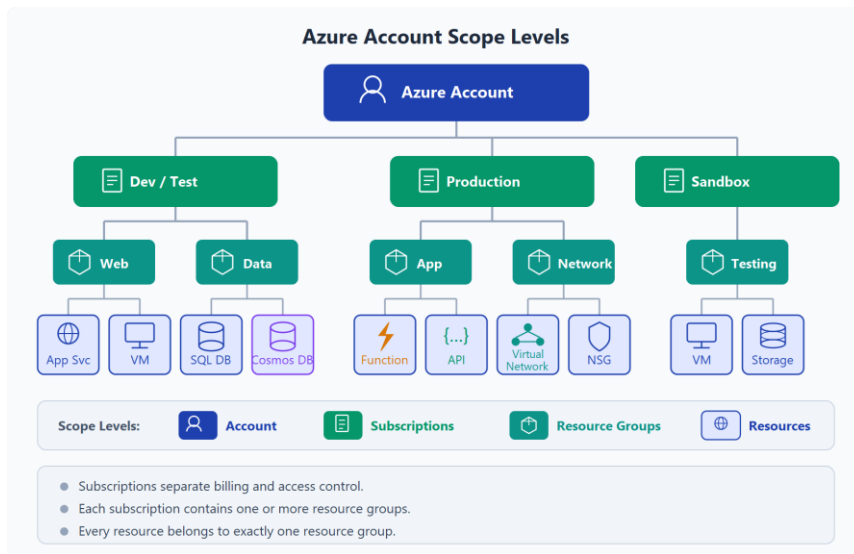
What is Microsoft Azure?

Microsoft Azure Service Categories				
Compute Virtual Machines App Service Azure Functions Container Apps Azure Kubernetes Service	Networking Virtual Network Load Balancer VPN Gateway ExpressRoute Azure DNS	Storage Blob Storage Azure Files Queue Storage Table Storage Managed Disks	Databases Azure SQL Azure Cosmos DB PostgreSQL Flex MySQL Flex Redis Cache	AI + ML Azure OpenAI AI Services Machine Learning Bot Service AI Search
Identity + Security Microsoft Entra ID Key Vault Defender for Cloud DDoS Protection Firewall	DevOps + Management Azure DevOps Azure Monitor ARM Templates Azure Policy Cost Management	IoT IoT Hub IoT Central Digital Twins Stream Analytics Event Hubs	Analytics Synapse Analytics Data Factory Databricks HDInsight Azure Data Explorer	Integration Logic Apps API Management Service Bus Event Grid Notification Hubs

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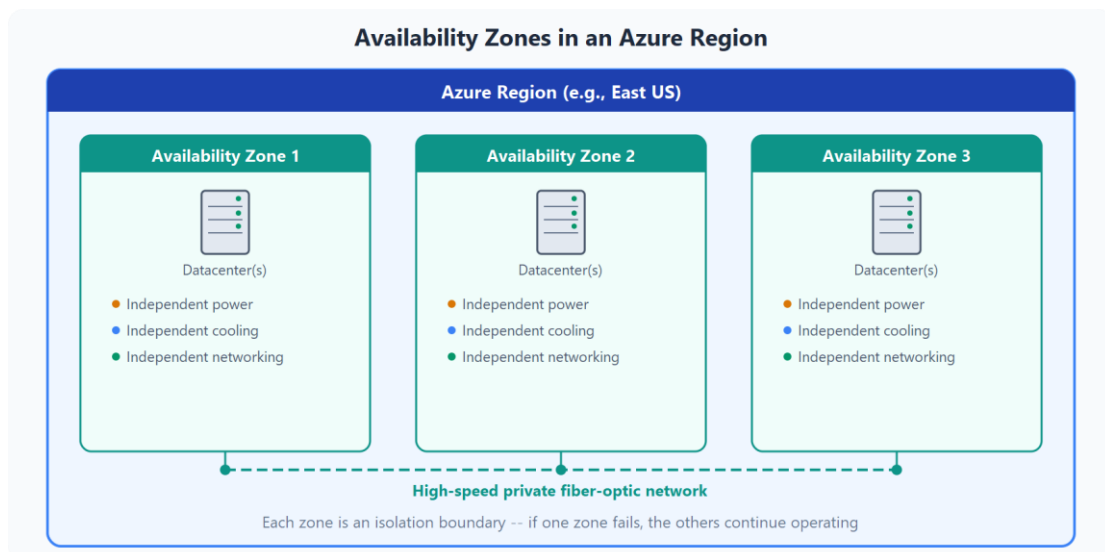
Get started with Azure accounts



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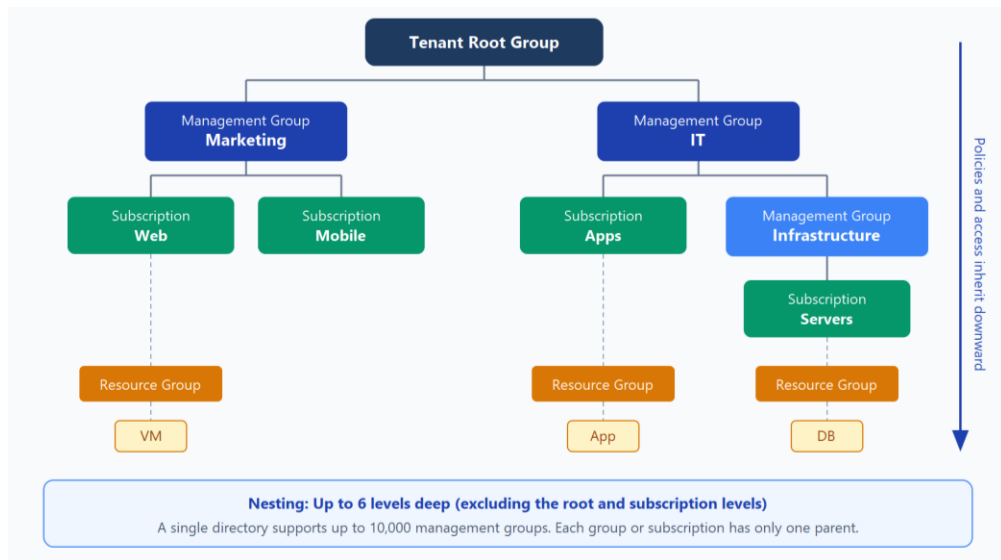
Azure physical infrastructure



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Azure management infrastructure



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Describe Azure compute services

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Decoding Azure Virtual Machine names

How to read Standard_D2s_v5



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Azure Virtual Machines families

VM size families

Choose the family that best matches workload behavior.



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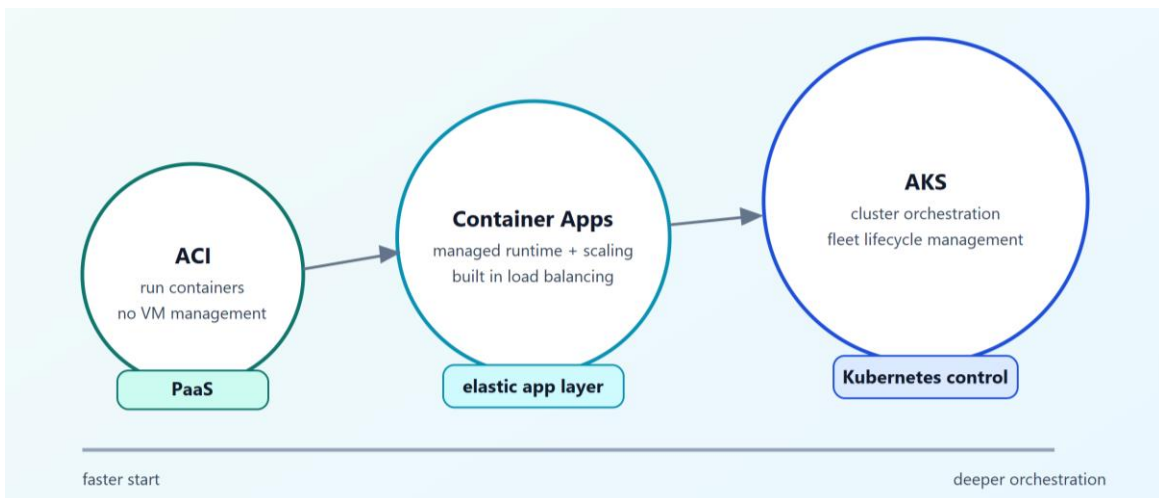
Azure Virtual Desktop



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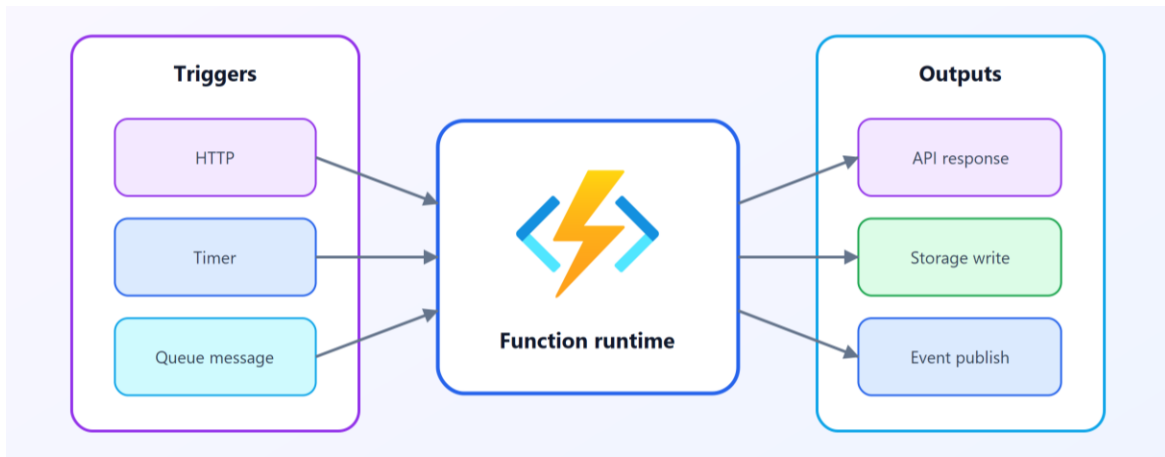
Azure container services



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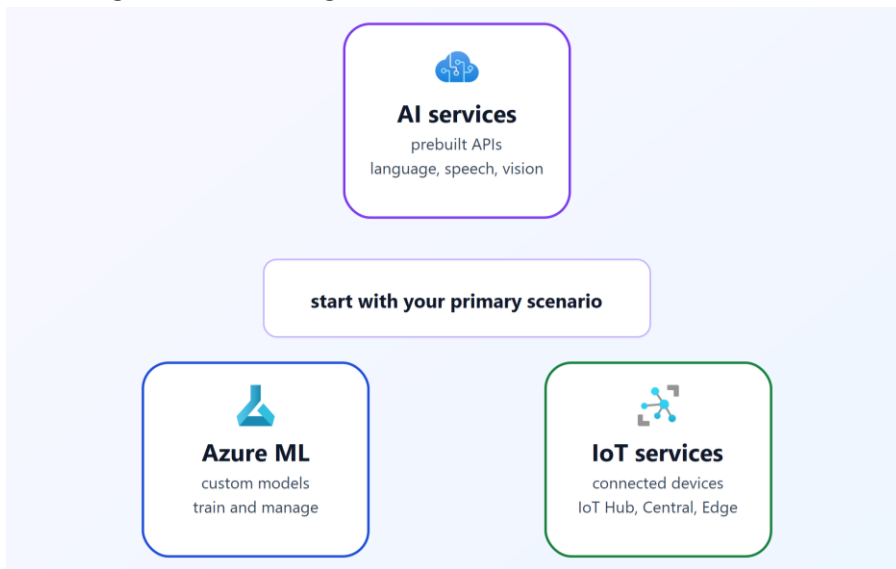
Azure Functions



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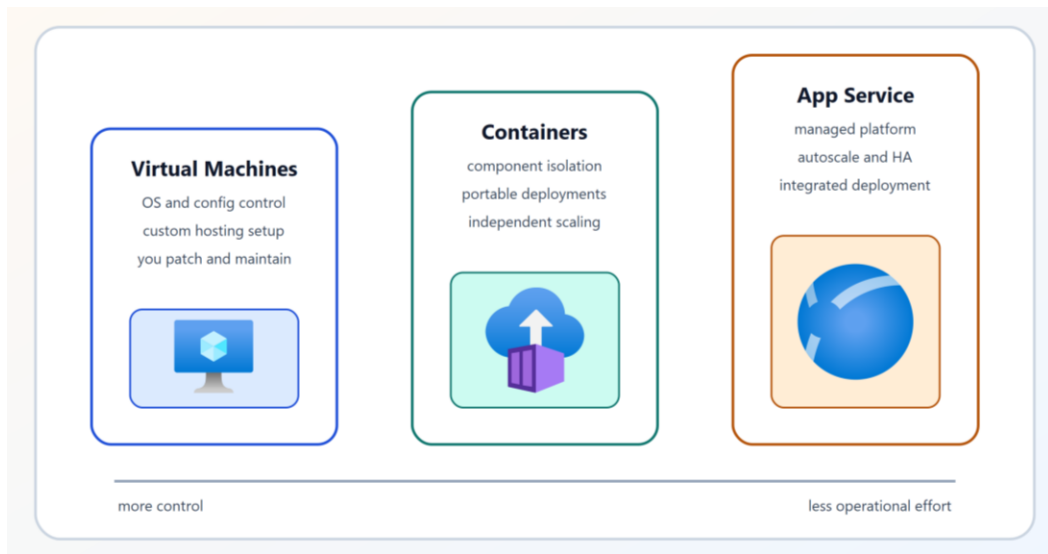
AI, machine learning, and IoT and Edge services



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Application hosting options



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Describe Azure networking services

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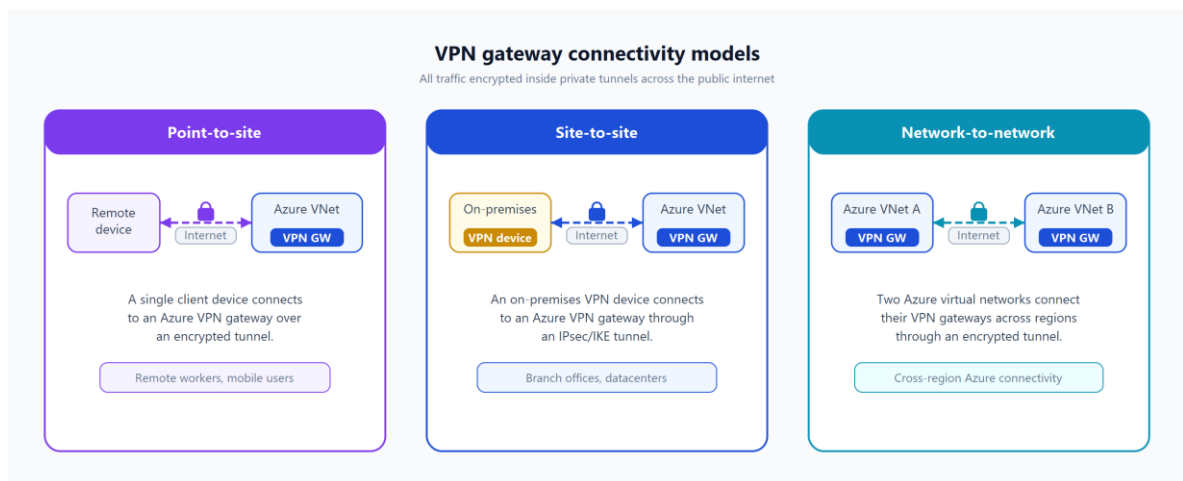
Azure virtual networks

- VNets provide isolation, segmentation, internet and cross-network connectivity
- Public endpoints have public IPs; private endpoints use IPs within the VNet address space
- Network Security Groups (NSGs) filter traffic with inbound/outbound rules by IP, port, and protocol
- VNet peering connects VNets (even cross-region) over the Microsoft private backbone
- On-premises connectivity: point-to-site VPN, site-to-site VPN, and Azure ExpressRoute

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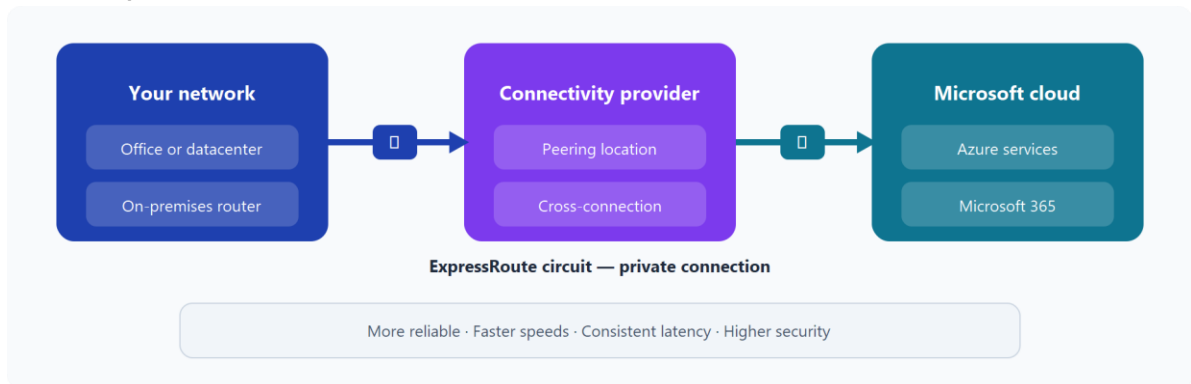
Azure VPN Gateway



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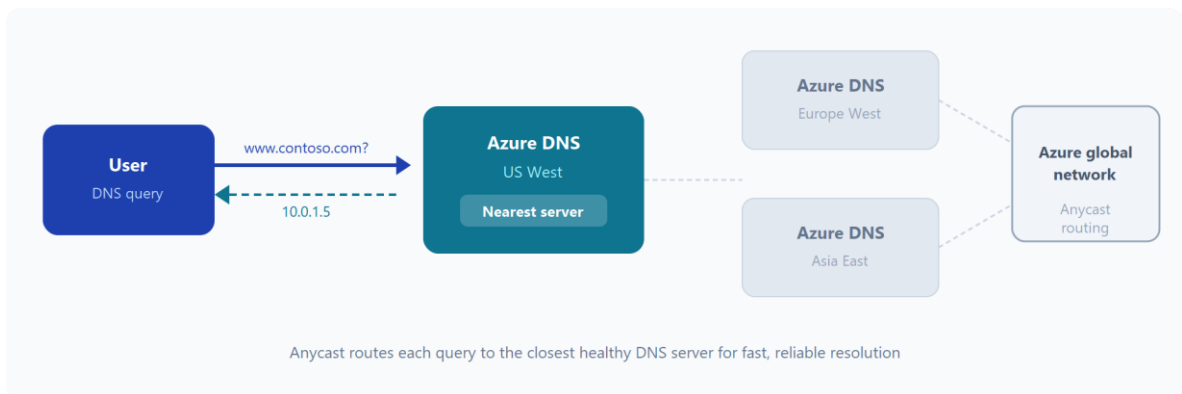
Azure ExpressRoute



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Azure DNS



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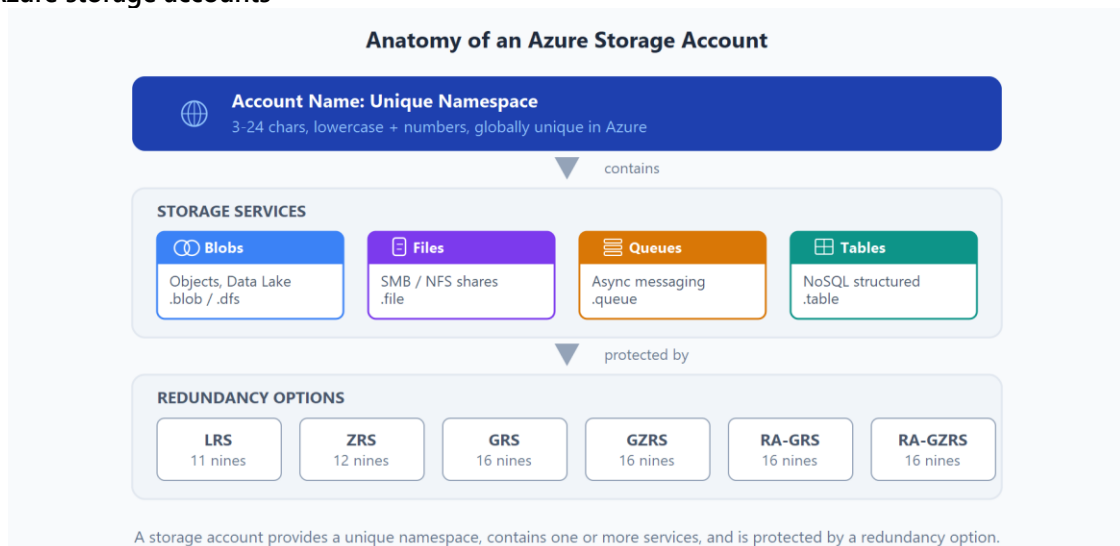
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Describe Azure storage services

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Azure storage accounts



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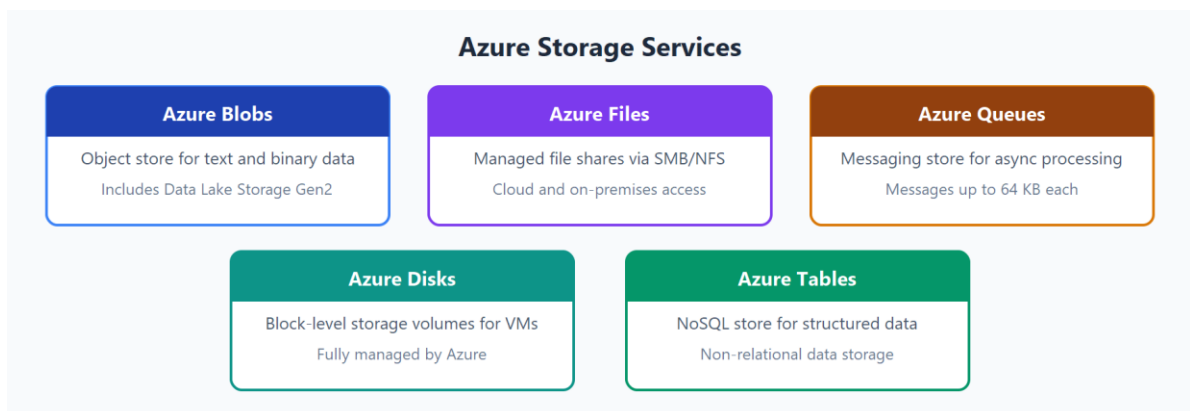
Azure storage redundancy

- LRS replicates data 3x in a single datacenter (11 nines durability); lowest cost
- ZRS replicates across 3 availability zones in the primary region (12 nines durability)
- GRS uses LRS in both primary and secondary regions (16 nines durability)
- RA-GRS and RA-GZRS enable read access to secondary region data before failover

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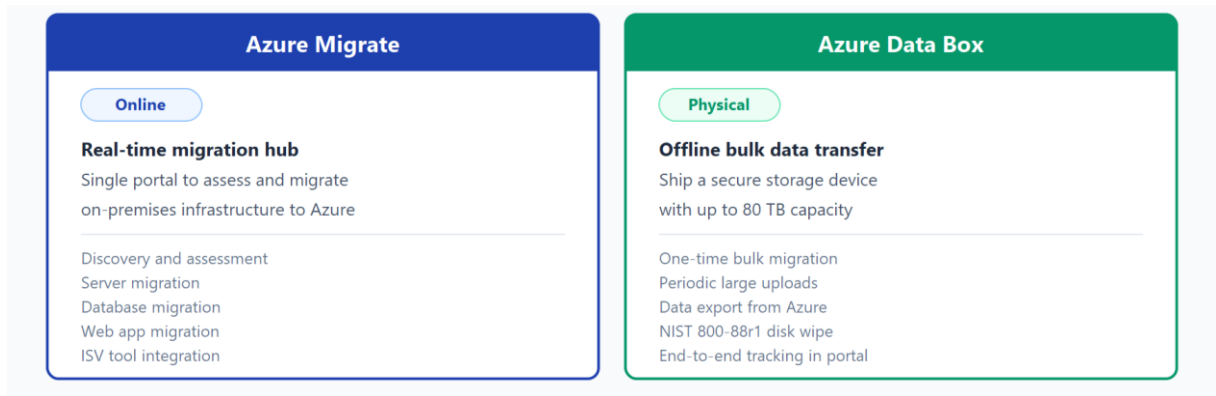
Azure storage services



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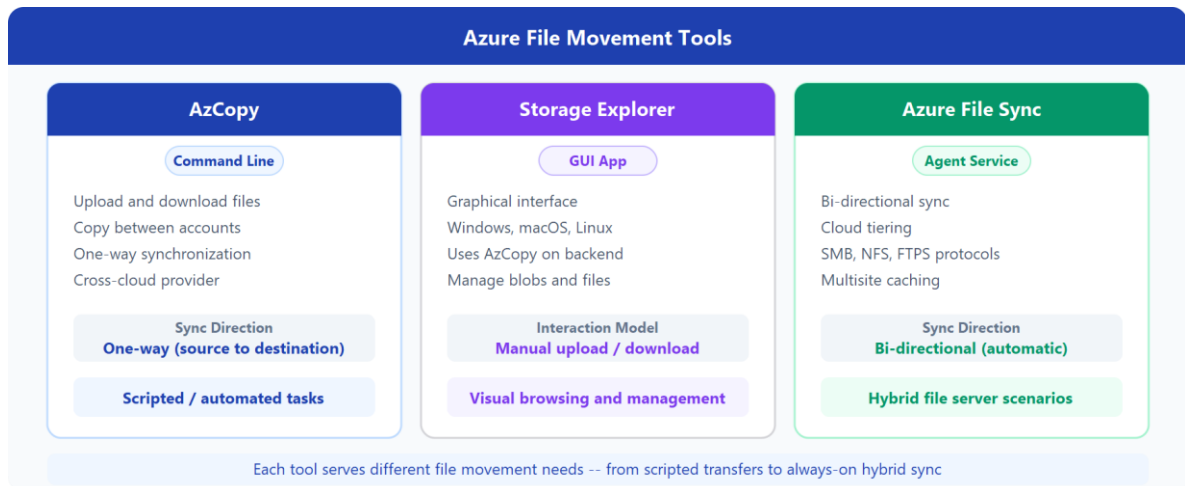
Azure data migration options



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Azure file movement options



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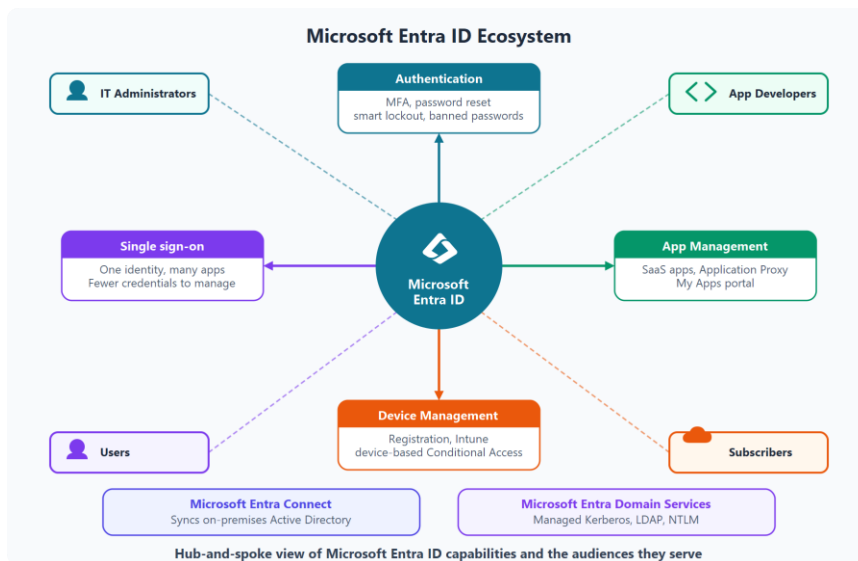
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Describe Azure identity, access, and security

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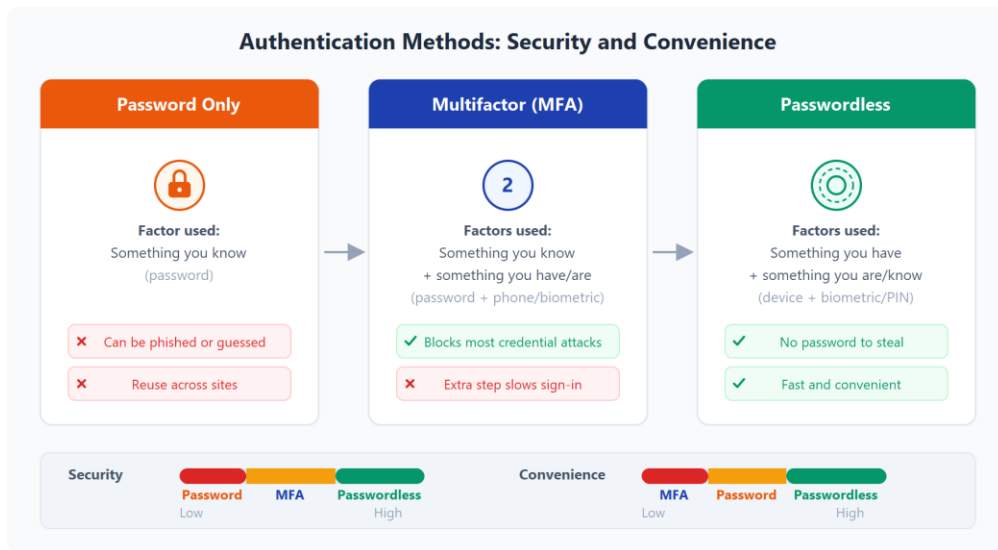
Microsoft Entra ID and Domain Services



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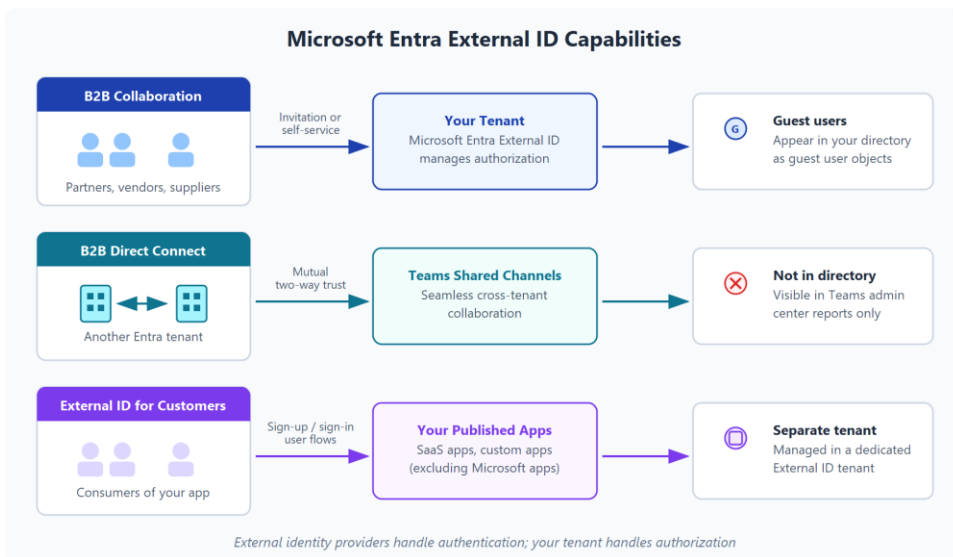
Authentication methods



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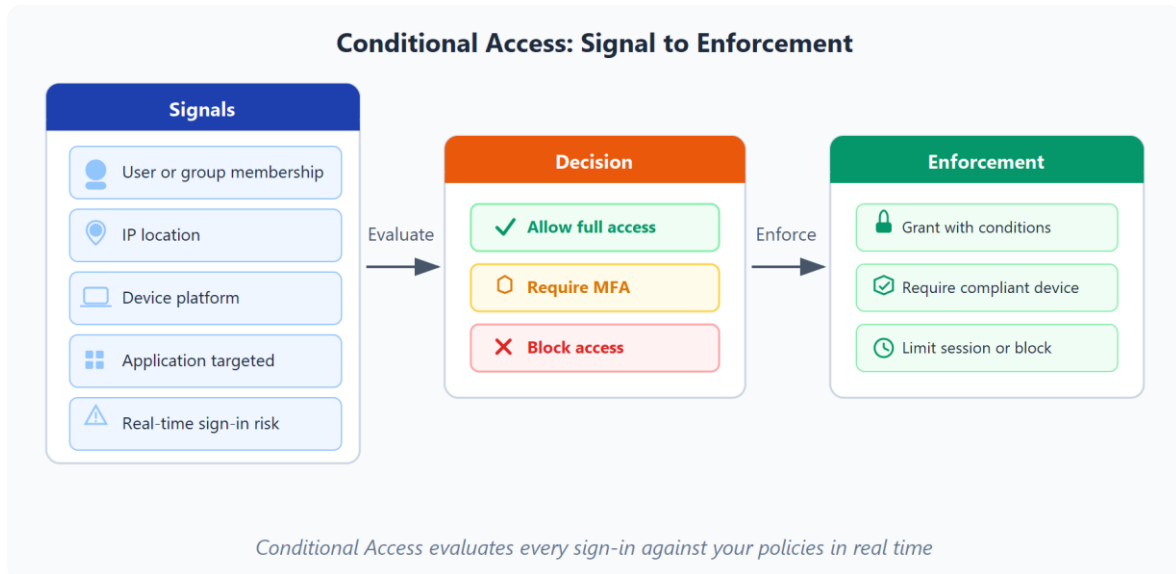
External identities



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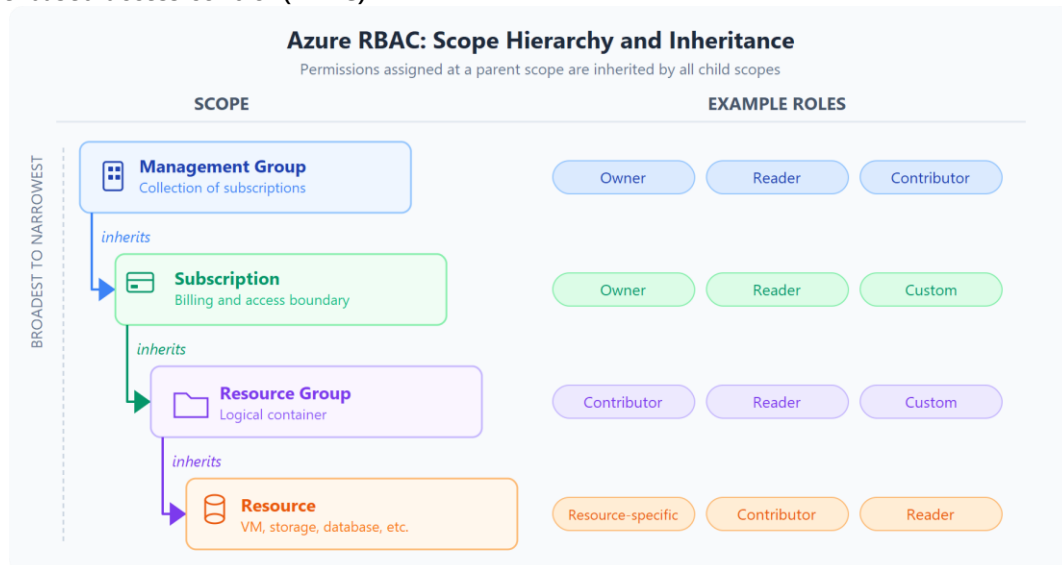
Conditional Access



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Role-based access control (RBAC)



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Zero Trust model



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Defense in depth



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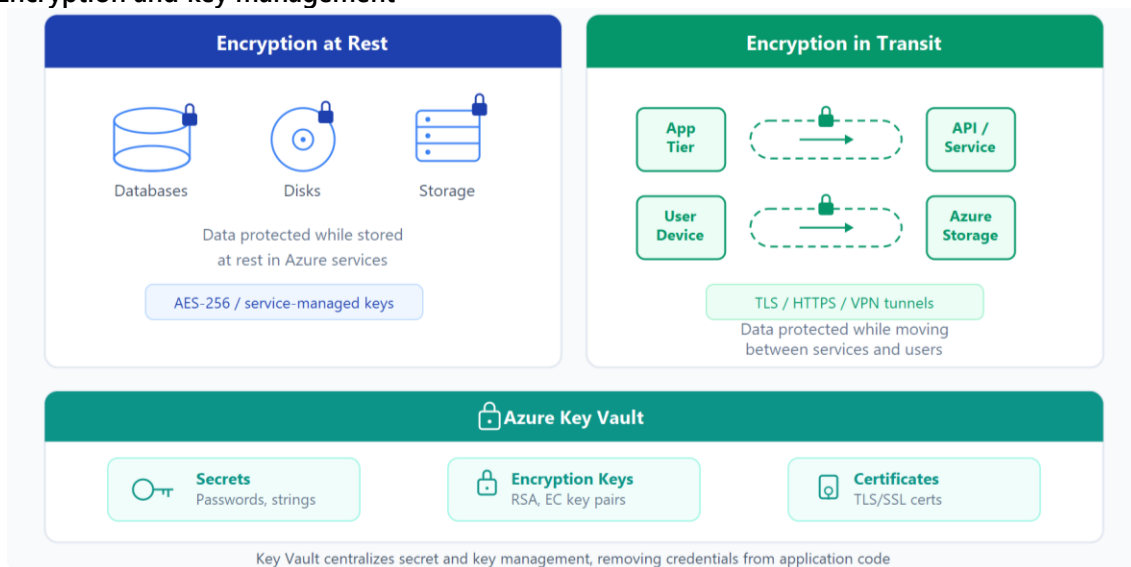
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Layer	Azure Services
Physical	Azure Datacenter Security
Identity	Microsoft Entra ID, MFA
Perimeter	Azure Firewall, DDoS Protection
Network	VNet, NSG, Private Link
Compute	Azure Defender, VM Security
Application	Key Vault, DevSecOps
Data	Encryption, Purview, DLP

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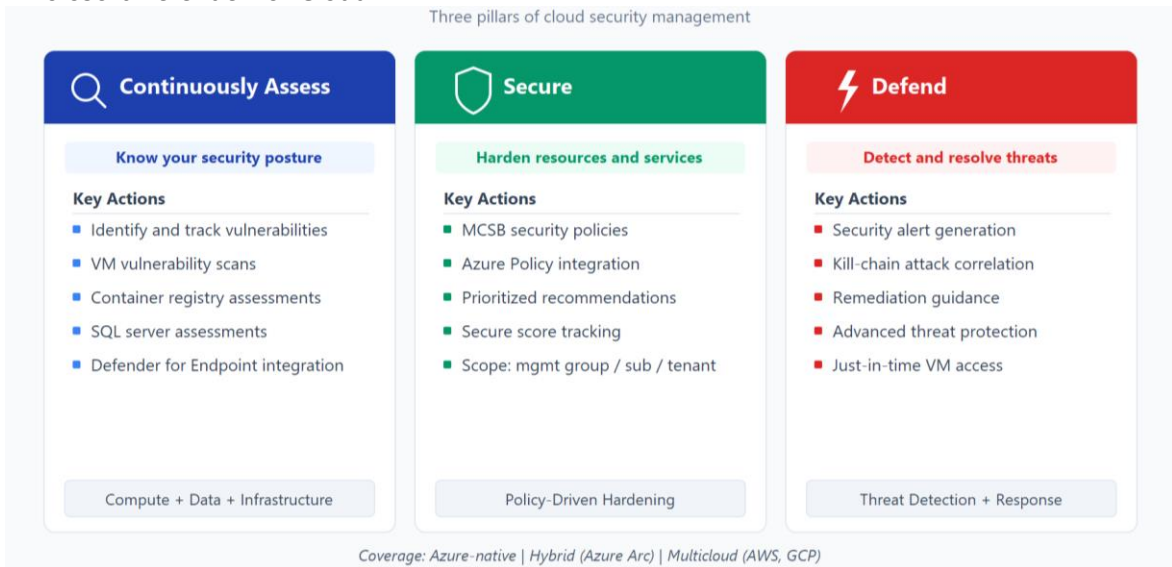
Encryption and key management



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Microsoft Defender for Cloud



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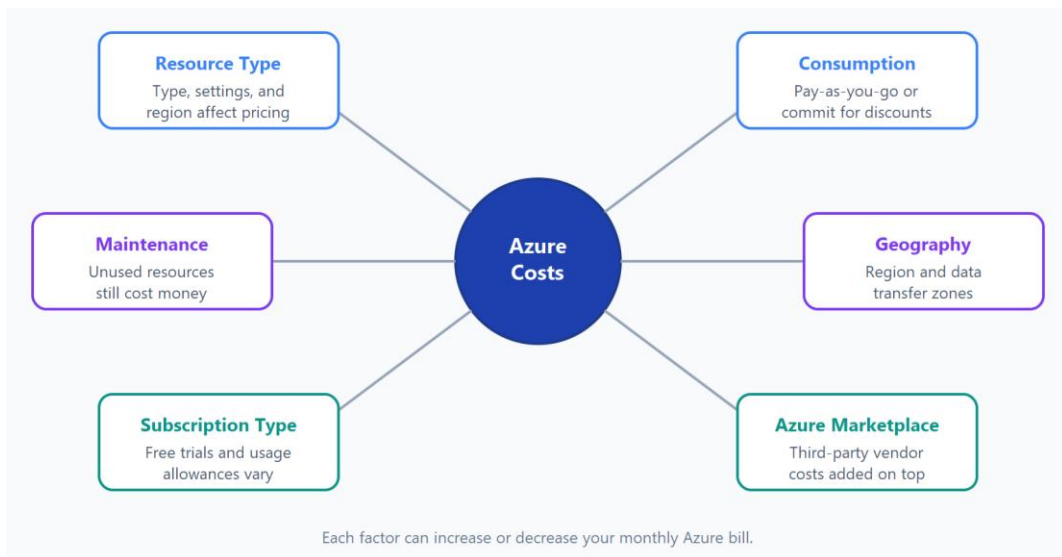
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Describe cost management in Azure

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Factors that affect costs in Azure



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ปัจจัย	ส่งผลต่อค่าใช้จ่ายอย่างไร
Resource Type	ขนาดและประเภทของบริการ
Consumption	ปริมาณการใช้งาน
Geography	Region และ Data Transfer
Azure Marketplace	บริการ Vendor ภายนอก
Subscription Type	ราคาและสิทธิประโยชน์
Maintenance	Resource ที่ถูกทิ้งไว้แต่ยังมีค่าใช้จ่าย

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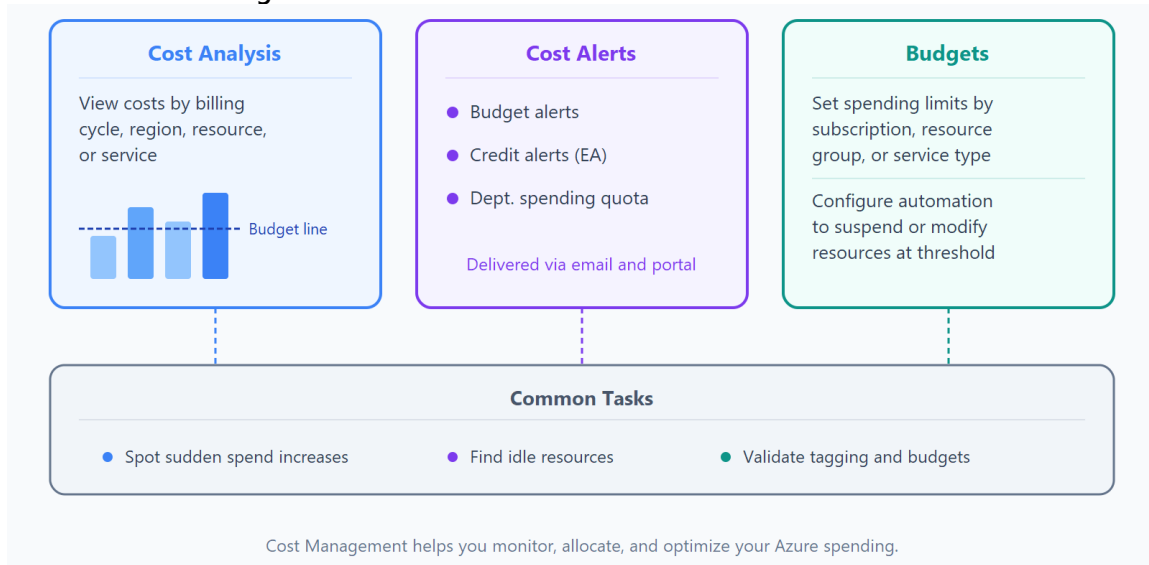
Pricing calculator

- Estimate costs before provisioning resources in Azure
- Build configurations for individual resources or full solutions
- Compare estimates across regions, tiers, and redundancy options
- Use estimates as a baseline for budget approval
- Revisit when architecture changes to keep planned spend aligned
- Information only: nothing is provisioned and no charges are incurred

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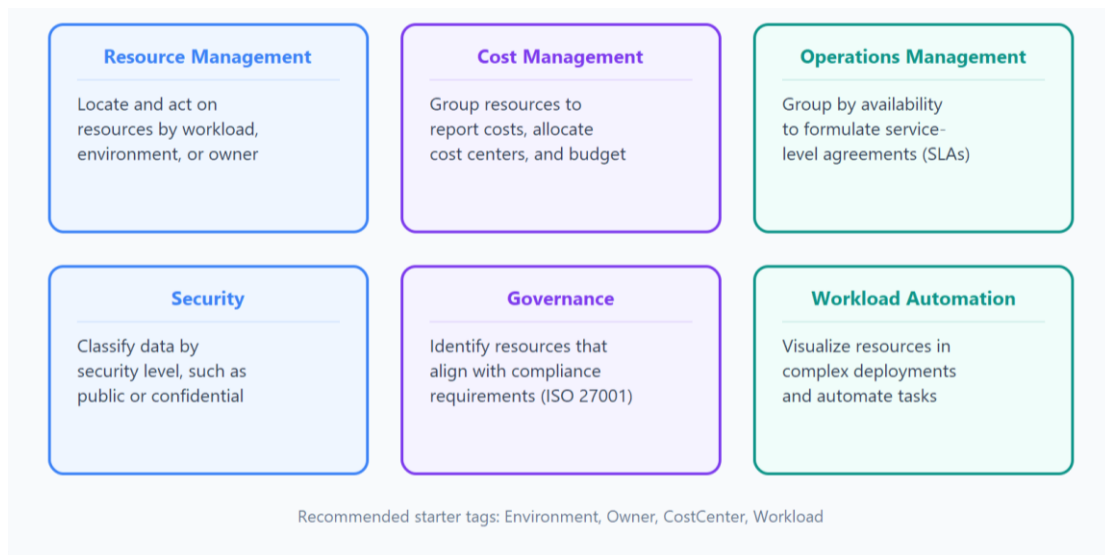
Microsoft Cost Management



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Purpose of resource tags



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Cost optimization options

- Reservations: commit to specific resources for 1 or 3 years at discounted pricing
- Savings plan for compute: commit to hourly spend amount across eligible compute services
- Spot Virtual Machines: use unused Azure capacity at reduced prices for interruptible workloads
- Decision guide: Reservations for stable workloads, savings plans for flexible compute, Spot for fault-tolerant jobs
- Review choices regularly as workload patterns change over time

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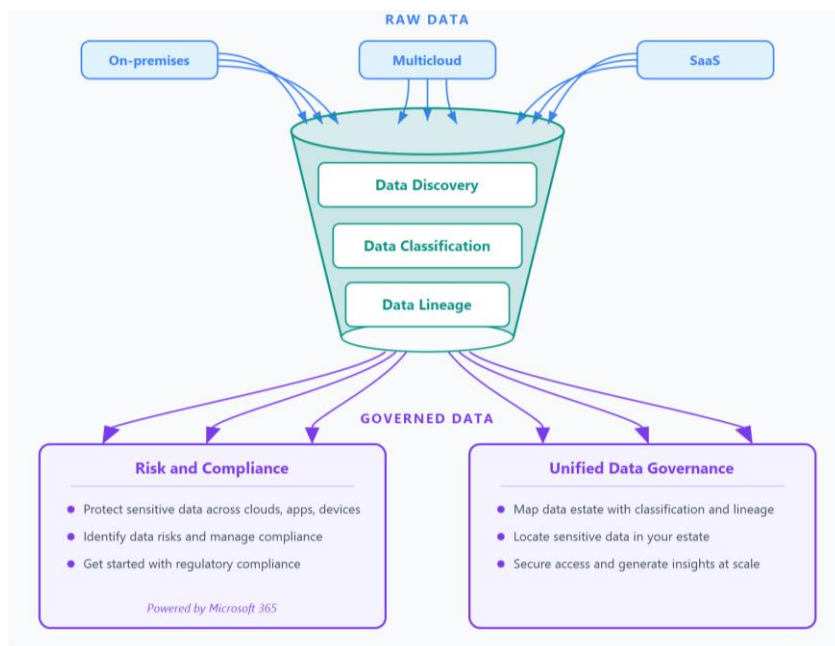
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Describe features and tools
for governance and
compliance

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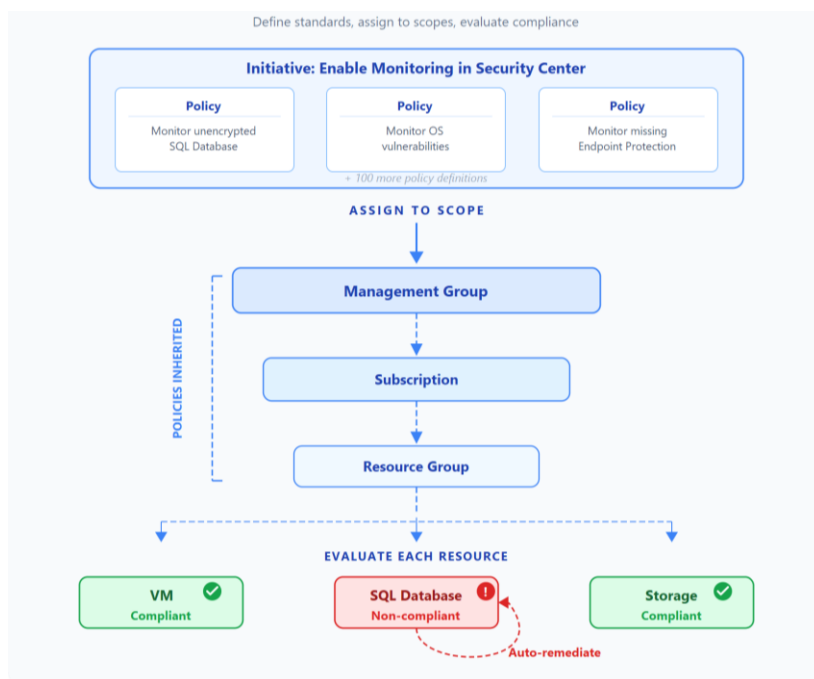
Microsoft Purview



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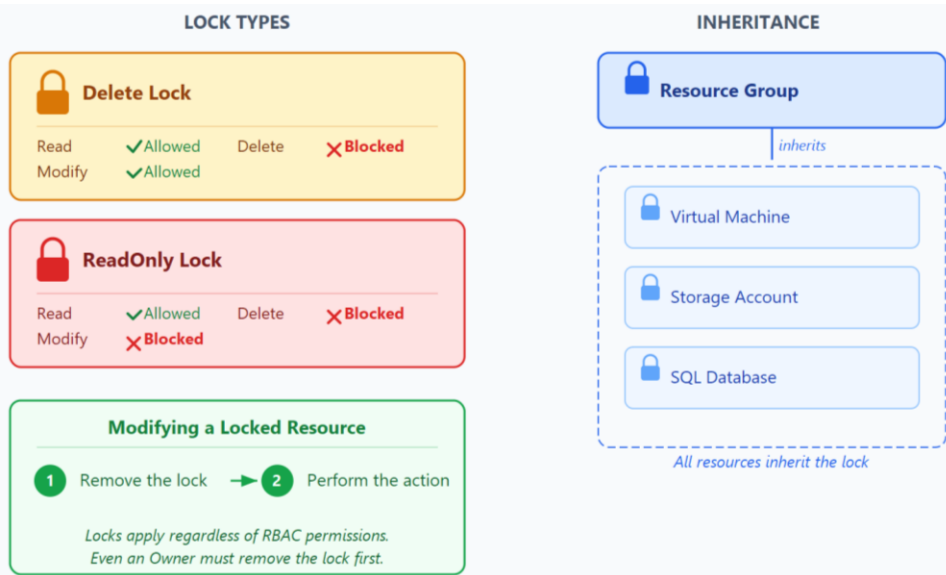
Azure Policy



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Resource locks



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Service Trust Portal

- Portal for accessing Microsoft security, privacy, and compliance resources
- Contains details about controls and processes that protect cloud services
- Requires authentication with a Microsoft Entra work or school account
- My Library: save and pin documents, set up notifications for updates
- All Documents: single landing place for all available compliance documents
- Reports and documents available for download for at least 12 months

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Describe features and tools for managing and deploying Azure resources

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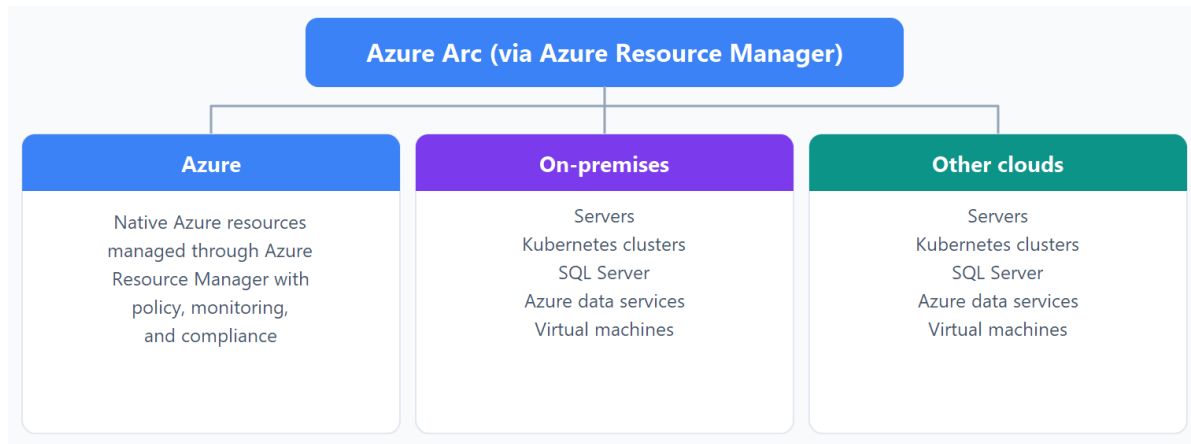
Tools for interacting with Azure

GRAPHICAL Azure portal	ENVIRONMENT Cloud Shell	COMMAND LINE Azure PowerShell	COMMAND LINE Azure CLI	AI-ASSISTED Copilot in Azure
Web-based console to build, manage, and monitor cloud resources	Browser-based shell with built-in auth. No local install required	Cmdlet-based scripting for repeatable automation tasks	Bash-based commands with same capabilities as PowerShell	AI assistant for contextual guidance in natural language

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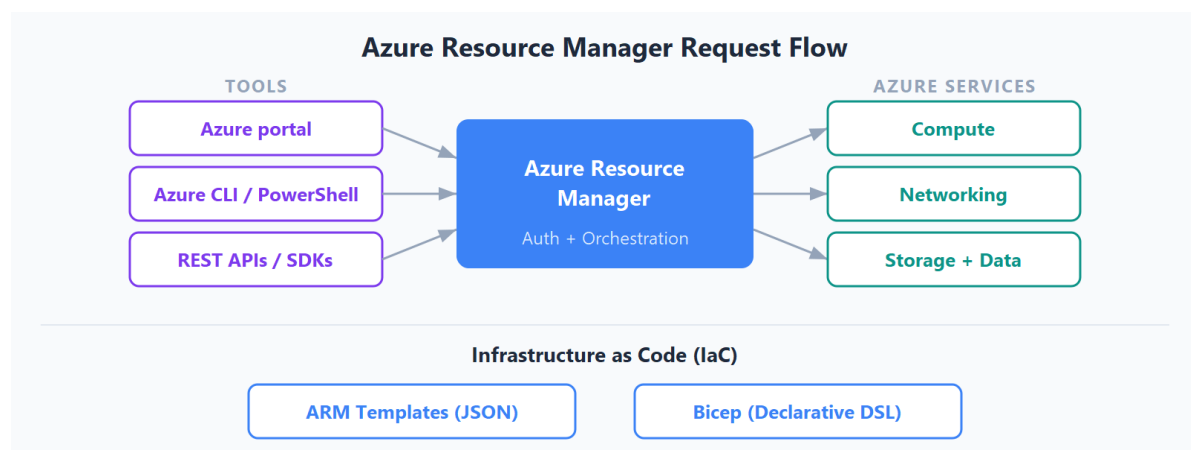
Azure Arc – Unified management across environments



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Azure Resource Manager, ARM templates, and Bicep



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Describe monitoring tools in Azure

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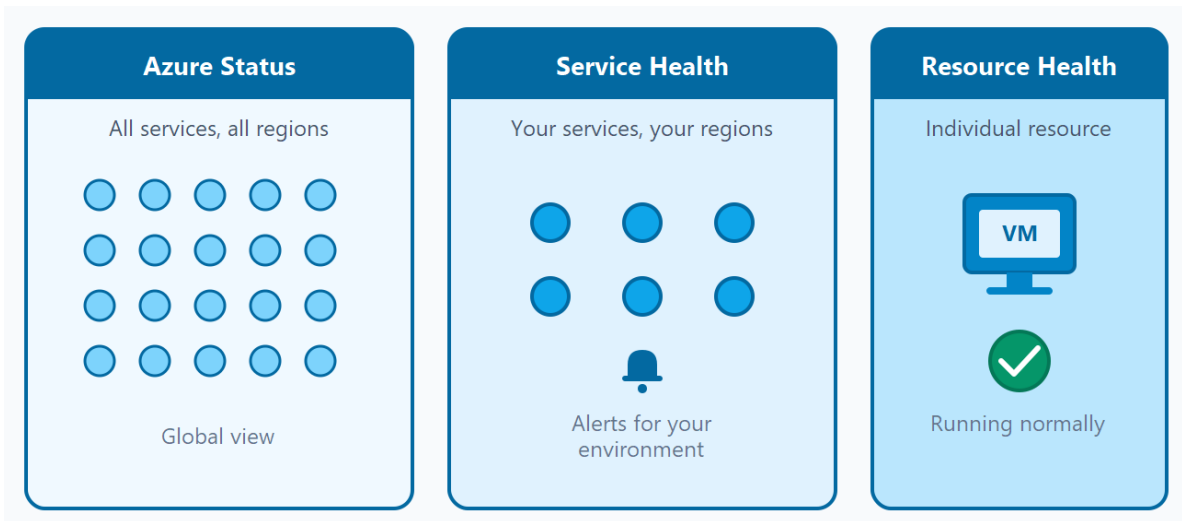
Azure Advisor

- Evaluates Azure resources and makes personalized best-practice recommendations
- Reliability: flags configuration risks to application availability
- Security: detects threats and vulnerabilities that could lead to breaches
- Performance: identifies changes that can speed up applications
- Operational Excellence: suggests workflow and deployment improvements
- Cost: finds ways to reduce Azure spending

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Azure Service Health



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Azure Monitor

- Platform for collecting, analyzing, and acting on telemetry across Azure, on-premises, and multicloud
- Log Analytics: write and run queries against collected logs and metrics
- Azure Monitor Alerts: notify when conditions are met with action groups for response
- Metric-based alerts (e.g., CPU above 80%) and log-based alerts (e.g., error patterns)
- Application Insights: monitors web app performance, request rates, response times, and failures
- Synthetic availability tests verify apps are responding during low-traffic periods

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Thank you for attending this course

Here are some reminders



Celebrate your new skills

- Redeem your achievement
- Share with us and your network



Let us know how we did

- Give us your feedback
- Survey will be sent via email



Become Microsoft Certified

- Explore additional resources to help prepare
- Schedule your exam



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What could be next in your learning journey?

Learning Path 1: Microsoft Azure AI fundamentals

- AI workloads and considerations
- Fundamental principles of machine learning on Azure
- Features of computer vision workloads on Azure
- Features of Natural Language Processing workloads on Azure
- [aka.ms/AI900](#)

Learning Path 2: Microsoft security, compliance, and identity fundamentals

- Concepts of security, compliance, and identity
- Capabilities of Microsoft Azure Active Directory, part of Microsoft Entra
- Capabilities of Microsoft security solutions
- Capabilities of Microsoft compliance solutions
- [aka.ms/SC900](#)

Learning Path 3: Microsoft Azure Data fundamentals

- Core data concepts
- Considerations for relational data on Azure
- Considerations for working with nonrelational data on Azure
- Analytics workloads on Azure
- [aka.ms/DP900](#)

Learning Path 4: Microsoft Azure Administrator

- Manage Azure identities and governance
- Implement and manage storage
- Deploy and manage Azure compute resources
- Implement and manage virtual networking
- Monitor and maintain Azure resources
- [aka.ms/AZ104](#)

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